Worksheet 5-5 & 5-2 Absolute Extrema #1
AP Calculus Name_____

1. The abs. min on $[-1,2]$ is 1 at $x=2$
The abs. max on $[-1,2]$ is 4 at $x=-1$

2. The abs min. on $[-3,1]$ is -13 at $x=-3$
The abs max on $[-3,1]$ is 5 at $x=0$

3. The abs min on $[0,6]$ is -8 at $x=2$
The abs max on $[0,6]$ is 24 at $x=6$

4. The abs min on $[-1,2]$ is 0 at $x=0$
The abs max on $[-1,2]$ is 14 at $x=2$

5. The abs min on $[-8,8]$ is -2 at $x=-8$
The abs max on $[-8,8]$ is 2 at $x=8$

6. The abs. min on $[-1,6]$ is $\frac{-1}{2}$ at $x=-1$
The abs. max on $[-1,6]$ is $\frac{2}{3}$ at $x=6$

7. The abs. min on $[0,2\pi]$ is -2 at $x=\frac{3\pi}{2}$
The abs. max on $[0,2\pi]$ is 2 at $x=\frac{\pi}{2}$

9. The minimum velocity on $0\le t\le 6$ is $\frac{-9}{2}$ at $t=\frac{7}{2}$

10. The max acceleration on $0\le t\le 3$ is 8 at $t=0$